

10/4/21

Mathematics

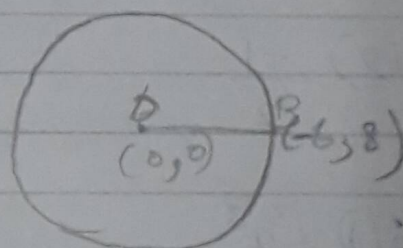
$$\begin{aligned}
 1. \quad p(x) &= x^2 - 2x - (7P + 3) \\
 p(-4) &= 16 - 2(-4) - (7P + 3) \\
 &= 16 + 8 - 7P - 3 \\
 0 &= 24 - 3 - 7P \\
 &= 21 - 7P \\
 7P &= 21 \\
 P &= 3
 \end{aligned}$$

(1) 3

2. (4) No solution.

3. Radius = OB

$$\begin{aligned}
 OB &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \\
 &= \sqrt{(6 - 0)^2 + (8 - 0)^2} \\
 &= \sqrt{6^2 + 8^2} \\
 &= \sqrt{36 + 64} \\
 &= \sqrt{100} \\
 &= 10 \text{ units}
 \end{aligned}$$



(1) 10 units.

4. (2) Proportional

$$\begin{aligned}
 5. \quad \sqrt{3} \tan \theta &= 2 \sin \theta \\
 \sqrt{3} \frac{\sin \theta}{\cos \theta} &= 2 \sin \theta \\
 \sqrt{3} \cdot 1 &= 2 \cos \theta \\
 \frac{\sqrt{3}}{2} &= \cos \theta
 \end{aligned}$$

(1)

$$\theta = 30^\circ$$

$$\sin^2 \theta - \cos^2 \theta$$

$$= \sin^2 30^\circ - \cos^2 30^\circ$$

$$= \left(\frac{1}{2}\right)^2 - \left(\frac{\sqrt{3}}{2}\right)^2$$

$$= \frac{1}{4} - \frac{3}{4}$$

$$= \frac{-2}{4} = \frac{-1}{2}$$

$$(2) \frac{-1}{2}$$

$$6(1) x^3 y^3$$

$$70 \quad 4x^2 - 2x + (k-4)$$

$$X+B = \frac{1}{2} \cdot XB$$

$$XB = \frac{2}{4} \left(\frac{-b}{a} \right)$$

$$= \frac{1}{2}$$

$$XB = \frac{c}{a}$$

$$= \frac{k-4}{4}$$

(2)

$$\begin{array}{r} 26 \cdot 27 \\ 512 \cdot 274 \\ 6 \cdot 4 \\ 12 \cdot 16 \end{array}$$

$$\begin{array}{l} 2^3 \cdot 3^3 = 4 \times 27 \\ 2^3 \cdot 3^2 = 8 \times 9 \\ = 72 \end{array}$$

$$\begin{array}{r} 2 \mid 108, 72 \\ 2 \mid 54, 36 \\ 3 \mid 27, 18 \\ 3 \mid 9, 6 \\ 3 \mid 3, 2 \\ 2 \mid 1, 1 \end{array}$$

$$\begin{array}{l} 2 \\ 3 \\ 3 \\ 3 \\ 2 \end{array} \begin{array}{l} x \\ y \\ z \\ w \\ v \end{array}$$

$$\frac{x+y}{2} = \frac{1 \times 8}{2}$$

$$\frac{1}{2} = \frac{1}{2} \left(\frac{k-4}{4} \right)$$

$$\frac{1}{2} = \frac{k-4}{8}$$

$$\frac{8}{2} = k-4$$

$$\begin{aligned} k-4 &= 4 \\ k &= 4+4 \\ &= 8 \end{aligned}$$

$$8. A(0,0); B(3,\sqrt{3}); C(3,x)$$

$$AB=BC=CA$$

$$\begin{aligned} AB &= \sqrt{(x_2-x_1)^2 + (y_2-y_1)^2} \\ &= \sqrt{(3-0)^2 + (\sqrt{3}-0)^2} \\ &= \sqrt{3^2 + 3} \\ &= \sqrt{9+3} \\ &= \sqrt{12} \end{aligned}$$

$$\begin{aligned} BC &= \sqrt{(3-3)^2 + (x-\sqrt{3})^2} \\ \sqrt{12} &= \sqrt{0 + x^2 - 2\sqrt{3}x + 3} \\ (\sqrt{12})^2 &= (x^2 - 2\sqrt{3}x + 3)^2 \\ 12 &= x^2 - 2\sqrt{3}x + 3 \\ x^2 - 2\sqrt{3}x - 9 &= 0 \end{aligned}$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{2\sqrt{3} \pm \sqrt{(2\sqrt{3})^2 - 4(1)(9)}}{2}$$

$$= \frac{2\sqrt{3} \pm \sqrt{12 + 36}}{2}$$

$$= \frac{2\sqrt{3} \pm \sqrt{48}}{2}$$

$$= \frac{2\sqrt{3} \pm 4\sqrt{3}}{2}$$

$$= \frac{2\sqrt{3} - 4\sqrt{3}}{2} \quad ; \quad \frac{2\sqrt{3} + 4\sqrt{3}}{2}$$

$$= \frac{-2\sqrt{3}}{2} \quad ; \quad \frac{6\sqrt{3}}{2}$$

$$= -\sqrt{3} \quad ; \quad = 3\sqrt{3}$$

$x = -\sqrt{3}$ is rejected.

$$x = 3\sqrt{3}$$

9. $2px + 3y = 7$
 $2x + y = 6$

$\frac{a_1}{a_2} \neq \frac{b_1}{b_2} \Rightarrow$ one solution

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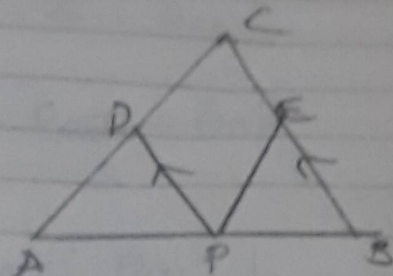
$$\frac{2p}{2} = \frac{3}{1}$$

$$p = 3$$

10. Given

$DP \parallel BC$

$$\frac{AD}{DC} = \frac{CE}{BE}$$



To Prove that
 $EP \parallel AC$

Statement	Reason
DP $DP \parallel CB$	Given
$\frac{AD}{DC} = \frac{AP}{PB}$ (1)	Basic Proportionality Theorem
$\frac{AD}{DC} = \frac{CE}{BE}$ (2)	Given
From (1) and (2)	
$\frac{AP}{PB} = \frac{CE}{BE}$	
Since $\frac{AP}{PB} = \frac{CE}{BE}$	Converse of Basic Proportionality Theorem
$EP \parallel AC$	

$$11. \tan \theta = \frac{1}{2}$$

$$\frac{\sin \theta}{\cos \theta} = \frac{1}{2}$$

$$\sin \theta = \frac{\cos \theta}{2}$$

$$\frac{1}{2} \sin \theta \cos \theta = \frac{1}{5}$$

$$\frac{1}{2} \times \frac{\cos \theta}{2} \times \cos \theta = \frac{1}{5}$$

$$\frac{\cos^2 \theta}{4} = \frac{1}{5}$$

$$\cos^2 \theta = \frac{4}{5}$$

$$\cos \theta = \frac{2}{\sqrt{5}}$$

$$\sin \theta = \frac{2}{\sqrt{5}} \times \frac{1}{2}$$

$$\sin \theta = \frac{1}{\sqrt{5}}$$

$$12. 5^4 \times 2^3 = 5000$$

$$n = 3$$

$$m = 4$$

TNFC

6

5000
5 | 1000
5 | 200
5 | 40
2 | 8
2 | 4
2 | 2
2 | 1

$$\frac{257.2}{5^4 \times 2^4}$$

$$= \frac{514}{10,000}$$

$$= 0.0514$$

13. ^s assume $\sqrt{3}$ is rational.

$$\sqrt{3} = \frac{a}{b} \text{ (a and b are coprime)}$$

~~$$\sqrt{3}b = a$$~~
~~$$3b^2 = a^2$$~~

$$3 = \frac{a^2}{b^2}$$

$$3b^2 = a^2$$

$$a = 3c \quad (1)$$

~~$$(3c)^2 = 3b^2$$~~

~~$$9c^2 = 3b^2$$~~

~~$$3c^2 = b^2$$~~

~~$$3c^2 = b^2$$~~

~~$$1b^2 = c^2$$~~

~~$$b^2 = c^2$$~~

~~$$b = c$$~~

~~$$b = c$$~~

~~$$b = c$$~~

~~$$b = c$$~~

~~$$b = c$$~~

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~~$$b = c$$~~

b is divisible by 3

Since a and b are divisible by 3. They are not co-prime. Hence $\sqrt{3}$ is

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an irrational number.

$$14 \cdot p(x) = 2x^2 - 3x + 1$$

$$\alpha + \beta = \frac{3}{2}$$

$$\alpha \cdot \beta = \frac{1}{2}$$

$$\frac{1}{2\alpha - 3} + \frac{1}{2\beta - 3}$$

$$\frac{2\beta - 3 + 2\alpha - 3}{4\alpha\beta - 6\alpha - 6\beta + 9}$$

$$= \frac{2(\alpha + \beta) - 6}{-2(3\alpha + 3\beta - 4\alpha\beta) + 9}$$

$$= \frac{2\left(\frac{3}{2}\right) - 6}{-2\left(3\left(\frac{3}{2}\right) - 4\left(\frac{1}{2}\right)\right) + 9}$$

$$= \frac{3 - 6}{-2\left(3\left(\frac{3}{2}\right) - 4\left(\frac{1}{2}\right)\right) + 9}$$

$$= \frac{-3}{-2(9 - 2) + 9}$$

$$= \frac{-3}{-14 + 9} = \frac{-3}{-5} = \frac{3}{5}$$

$$= \frac{-3}{-14 + 9} = \frac{-3}{-5} = \frac{3}{5}$$

⑧

$$\begin{aligned} 3x - (a+1)y &= -(2b+1) \\ 5x + (1-2a)y &= 3b \end{aligned}$$

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} \Rightarrow \text{infinitely many solutions}$$

$$\frac{3}{5} = \frac{-a-1}{1-2a} = \frac{2b-1}{3b}$$

$$\frac{-a-1}{1-2a} = \frac{3}{5}$$

$$-5(a+1) = 3(1-2a)$$

$$-5a - 5 = 3 - 6a$$

$$-5a + 6a = 3 + 5$$

$$a = 8$$

$$\frac{2b-1}{3b} = \frac{3}{5}$$

$$5(2b-1) = 3(3b)$$

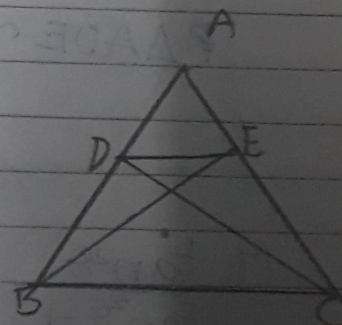
$$10b - 5 = 9b$$

$$10b - 9b = 5$$

$$b = 5$$

16. Given
 $\triangle ABE \cong \triangle ACD$

To Prove that
 $\triangle ADE \sim \triangle ABC$



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Statement	Reason
$AB = AC$ $AB - DB = AC - AE$ $AD = AE$	Given
$\frac{AD}{AB} = \frac{AE}{AC}$ (1)	If equals are subtracted from equal, then whole are equal. Basic Proportionality Theorem
$\frac{DB}{AB} = \frac{EC}{AC}$ (2)	Basic Proportionality Theorem
$\frac{DB}{AB} = \frac{AD}{AB}$	
$\frac{AE}{AC} = \frac{EC}{AC}$	
$\frac{AD}{DB} = \frac{AE}{EC}$	Basic Proportionality Theorem
$DE \parallel BC$	
$\angle A = \angle A$	Common angle
$\triangle ADE \sim \triangle ABC$	By SAS similarity criteria

$$17. \frac{\tan^2 \theta}{\tan^2 \theta - 1} + \frac{\operatorname{cosec}^2 \theta}{\sec^2 \theta - \operatorname{cosec}^2 \theta} = \frac{1}{\sin^2 \theta - \cos^2 \theta}$$

$$\frac{\sin^2 \theta \times 1}{\cos^2 \theta \times \sin^2 \theta - 1} + \frac{\cos \theta \times 1}{\sin^2 \theta \times \frac{1}{\cos^2 \theta} - \frac{1}{\sin^2 \theta}}$$

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$$\frac{\sin^2 \theta}{\cos^2 \theta} + 1 = \frac{\sin^2 \theta + \cos^2 \theta}{\cos^2 \theta}$$

$$\frac{\sin^2 \theta + \cos^2 \theta}{\sin^2 \theta - \cos^2 \theta}$$

$$\frac{1}{\sin^2 \theta - \cos^2 \theta} = \frac{1}{\sin^2 \theta - \cos^2 \theta}$$

LHS = RHS

18.

$$\begin{array}{ccc} & 3 & 2 \\ \text{A} & \text{C} & \text{B} \\ (-3, 7) & (1, 1) & (x, y) \end{array}$$

$$B = \left(\frac{x_2 m_1 + x_1 m_2}{m_1 + m_2}, \frac{y_2 m_1 + y_1 m_2}{m_1 + m_2} \right)$$

$$= \left(\frac{1 \times 3 + (-2) \times 2}{3 + 2}, \frac{1 \times 3 + 7 \times 2}{3 + 2} \right)$$

$$= \left(\frac{3 - 4}{5}, \frac{3 + 14}{5} \right)$$

$$= \left(\frac{-1}{5}, \frac{17}{5} \right)$$

19. $\frac{2x}{y+1} = 1$

$$2x = y + 1$$

$$2x - y = 1 \quad (1)$$

(11)

$$\frac{x+4}{2y} = \frac{1}{2}$$

$$x+4 = y$$

$$x - y = -4 \text{ (2)}$$

$$2x - y = 1$$

$$x - y = -4$$

$$x = 5$$

$$-y = -4 - 5$$

$$y = 9$$

$$y = 9$$

$$\frac{x}{y} = \frac{5}{9}$$

$$21. \frac{\sin 30^\circ + 2 \cos 45^\circ + \tan^2 60^\circ}{1 \times \cot 45^\circ + \cos^2 30^\circ + \tan^2 45^\circ}$$

$$\frac{1 \times 1 + (\frac{\sqrt{3}}{2})^2 + 12}{2}$$

$$= \frac{1 + 2 \times 1 + \sqrt{3}}{2}$$

$$= \frac{1 + 2\sqrt{2} + 2\sqrt{3}}{2}$$

$$= \frac{1 + 2\sqrt{2} + 2\sqrt{3}}{2}$$

$$\frac{(1+2\sqrt{2}+2\sqrt{3}) \times 2}{9}$$

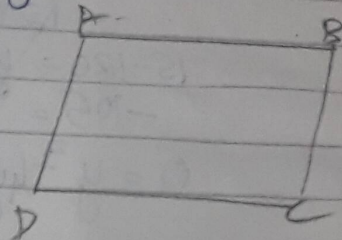
$$= \frac{2(1+2\sqrt{2}+2\sqrt{3})}{9}$$

$$\frac{a+b}{a \times \frac{c}{b}}$$

22. A(1,2); B(2,3); C(6,5); D(x,y)

$$AB = DC$$

$$AD = BC$$



$$\begin{aligned} AB &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \\ &= \sqrt{(2-1)^2 + (3-2)^2} \\ &= \sqrt{1+1} \\ &= \sqrt{2} \end{aligned}$$

$$\begin{aligned} DC &= \sqrt{(x-6)^2 + (y-5)^2} \\ \sqrt{2} &= \sqrt{x^2 - 12x + 36 + y^2 - 10y + 25} \\ (\sqrt{2})^2 &= (\sqrt{x^2 + y^2 - 12x - 10y + 61})^2 \\ 2 &= x^2 + y^2 - 12x - 10y + 61 \\ -59 &= x^2 - 12x + y^2 - 10y \quad (1) \end{aligned}$$

$$\begin{aligned} BC &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \\ &= \sqrt{(6-2)^2 + (5-3)^2} \\ &= \sqrt{4^2 + 2^2} \\ &= \sqrt{16+4} \\ &= \sqrt{20} \end{aligned}$$

$$\begin{aligned} AD &= \sqrt{(x-1)^2 + (y-2)^2} \\ (\sqrt{20})^2 &= (\sqrt{x^2 - 2x + 1 + y^2 - 4y + 4})^2 \\ 20 &= x^2 - 2x + y^2 - 4y + 5 \\ 15 &= x^2 - 2x + y^2 - 4y \quad (2) \end{aligned}$$

(13)

$$-59 = x^2 - 12x + y^2 - 10y$$

$$= x(x-12) + y(y-10)$$

$$-59 = (x+y)(x-12) + y(y-10)$$

$$x=12 \rightarrow x=y$$

$$x=-y$$

$$15 = 12^2 - 2(12) + y^2 - 4y$$

$$= 144 - 24 + y^2 - 4y$$

$$= 120 + y^2 - 4y$$

$$15 - 120 = y^2 - 4y$$

$$-105 = y^2 - 4y$$

$$0 = y^2 - 4y + 105$$

$$y = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{4 \pm \sqrt{4^2 - 4(1)(105)}}{2}$$

$$= \frac{4 \pm \sqrt{16 - 420}}{2}$$

$$= \frac{4 \pm \sqrt{404}}{2}$$

$$= \frac{4 \pm \sqrt{4 \times 101}}{2}$$

$$= \frac{4 \pm 4\sqrt{101}}{2}$$

$$= \frac{4 - 4\sqrt{101}}{2}$$

$$= \frac{4 + 4\sqrt{101}}{2}$$

(14)

$$= \frac{2(1-\sqrt{101})}{2} > \frac{2(1+\sqrt{101})}{2}$$

$$= 2(1-\sqrt{101}) > 2(1+\sqrt{101})$$

~~Q. 12~~ $x = 12$
 $y = -12$

23. ~~Q. 23~~ Circumference = 1980m

$$A = 330 \text{ m/min}$$

$$B = 198 \text{ m/min}$$

$$C = 220 \text{ m/min}$$

$$D = 1980$$

~~Q. 23~~

$$S = \frac{D}{T}$$

$$T = \frac{D}{S}$$

$$T_1 = \frac{1980}{330} = 6 \text{ min}$$

$$T_2 = \frac{1980}{198} = 10 \text{ min}$$

$$T_3 = \frac{1980}{220} = 9 \text{ min}$$

HCF
 LCM of 6, 10, 9

$$\frac{33 \times 6}{198}$$

$$\frac{33 \times 6}{198}$$

$$\frac{33 \times 6}{198}$$

$$\frac{33 \times 6}{198}$$

$$\frac{33 \times 6}{198}$$

$$\frac{33 \times 6}{198}$$

$$\frac{33 \times 6}{198}$$

~~$$\begin{array}{r}
 2 \mid 6, 10, 9 \\
 3 \mid 8, 5, 9 \\
 5 \mid 1, 5, 3 \\
 3 \mid 1, 1, 3 \\
 1, 1, 1
 \end{array}$$~~

~~$$2 \times 3^2 \times 5 = 10 \times 9 = 90 \text{ mins}$$~~

$$\begin{array}{r}
 2 \mid 6 \\
 3 \mid 3 \\
 1
 \end{array}$$

~~$$\begin{array}{r}
 2 \mid 10 \\
 5 \mid 5 \\
 1
 \end{array}$$~~

$$\begin{array}{r}
 3 \mid 9 \\
 3 \mid 3 \\
 1
 \end{array}$$

$$\begin{array}{r}
 2 \mid 6, 10, 9 \\
 3 \mid 3, 5, 9 \\
 5 \mid 1, 5, 3 \\
 3 \mid 1, 1, 3 \\
 1, 1, 1
 \end{array}$$

90 mins = They will meet after 90 mins at starting point.

Q4. $2x + y = 70$
 $2y + x = 95$

$y = \text{Father}$
 $x = \text{Son}$

$$\begin{array}{r}
 95 \\
 2 \\
 \hline
 190 \\
 70 \\
 \hline
 95 \\
 20 \\
 \hline
 115
 \end{array}$$

$$\begin{array}{r}
 x + 2y = 95 \\
 (2x + y = 70) \times 2 \quad 2x + 4y = 140 \\
 \hline
 2x + y = 70 \\
 \hline
 3y = 120 \\
 y = 40
 \end{array}$$

$x = 15$
 Father = 40, Son = 15

(16)

25. $A(1, -1)$

m_1
 $C(2, 0)$

m_2

$B(6, 4)$

$$AC = \left(\frac{x_1 m_2 + x_2 m_1}{m_1 + m_2}, \frac{y_1 m_2 + y_2 m_1}{m_1 + m_2} \right)$$

$$= \left(\frac{1 \times m_2 + 2 \times m_1}{m_1 + m_2}, \frac{-1 \times m_2 + 0}{m_1 + m_2} \right)$$

$$= \left(\frac{m_2 + 2m_1}{m_1 + m_2}, \frac{-m_2}{m_1 + m_2} \right)$$

$CB = \dots$

26. $\operatorname{cosec} \theta - \sin \theta = m$
 $\sec \theta - \cos \theta = n$

$$\frac{1}{\sin \theta} - \sin \theta = m$$

$$\frac{1 - \sin^2 \theta}{\sin \theta} = m$$

$$\frac{\cos^2 \theta}{\sin \theta} = m$$

$$\frac{1}{\cos \theta} - \cos \theta = n$$

$$\frac{1 - \cos^2 \theta}{\cos \theta} = n \quad n = \frac{\sin^2 \theta}{\cos \theta}$$

$$(m^2n)^{\frac{2}{3}} + (mn^2)^{\frac{2}{3}}$$

$$\left(\frac{\cos^4\theta \times \sin^2\theta}{\sin^2\theta \cos^2\theta}\right)^{\frac{2}{3}} + \left(\frac{\cos^2\theta \times \sin^4\theta}{\sin^2\theta \cos^2\theta}\right)^{\frac{2}{3}}$$

$$\cos^2\theta^{\frac{2}{3}} + \sin^2\theta^{\frac{2}{3}}$$

$$\cos^2\theta + \sin^2\theta$$

$$1 = 1$$

$$\text{LHS} = \text{RHS}$$

Hence Proved.

$$27) 4.26 \text{ miles.}$$

$$(iii) 18.2.$$

$$(iv) \text{ Hotel B to mountain} = 5 \text{ miles.}$$

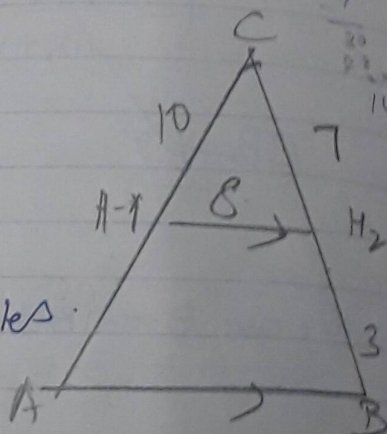
$$3x + 7x = 5$$

$$10x = 5$$

$$x = \frac{5}{10}$$

$$= \frac{1}{2}$$

$$\frac{1}{2} = 3.5 \text{ miles.}$$



$$28) P(2,5), Q(x,y), R(8,3)$$

$$m_1 : m_2$$

$$1 : 2$$

$$PQ = \left(\frac{x_1 m_2 + x_2 m_1}{m_1 + m_2}, \frac{y_1 m_2 + y_2 m_1}{m_1 + m_2} \right)$$

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$$= \left(\frac{2 \times 2 + 5 \times 1}{1+2}, \frac{5 \times 2 + y}{1+2} \right)$$

$$= \left(\frac{4+5}{3}, \frac{10+y}{3} \right)$$

$$= \left(\frac{9}{3}, \frac{10+y}{3} \right)$$

$$= \left(3, \frac{10+y}{3} \right)$$

$$x=3$$

$$QR = \left(\frac{x_1 m_2 + x_2 m_1}{m_1 + m_2}, \frac{y_1 m_2 + y_2 m_1}{m_1 + m_2} \right)$$

$$= \left(\frac{3 \times 2 + 8 \times 1}{1+2}, \frac{3 \times 1 + y \times 2}{1+2} \right)$$

$$= \left(\frac{6+8}{3}, \frac{3+2y}{3} \right)$$

=

$$Q = (4, 4)$$

$$\begin{aligned} \text{(ii)} \quad PR &= \sqrt{(8-2)^2 + (3-5)^2} \\ &= \sqrt{6^2 + 2^2} \\ &= \sqrt{36+4} \\ &= \sqrt{40} \end{aligned}$$

$$\text{(iii)} \quad (3, 4)$$

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

$$= \left(\frac{10}{2}, \frac{8}{2} \right)$$

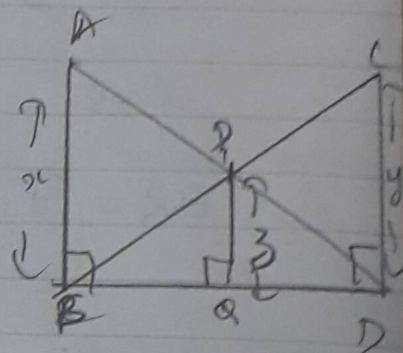
$$= (5, 4)$$

~~29. A = B = 100 km~~

30. Given - $\angle ABD = \angle CDB = \angle PQB = 90^\circ$

To Prove that

$$\frac{1}{x} + \frac{1}{y} = \frac{1}{z}$$



Statement	Reason
$\angle ABD = \angle CDB = 90^\circ$ (R)	Given
$CD = CD$ (S)	Common side
$AD = CB$ (H)	
$\triangle ABD \cong \triangle CBD$	RHS $\cong$

$$x = y$$

$$\frac{1}{x} + \frac{1}{y} = \frac{1}{z}$$